THE JOURNAL OF BIOLOGICAL CHEMISTRY
Vol. 258, No. 4, Issue of February 25, pp. 2508-2513, 1983
Printed in U.S.A.

Apolipoprotein A-I$_{Milano}$

DETECTION OF NORMAL A-I IN AFFECTED SUBJECTS AND EVIDENCE FOR A CYSTEINE FOR ARGININE SUBSTITUTION IN THE VARIANT A-I*

(Received for publication, August 30, 1982)

Karl H. Weisgraber, Stanley C. Rall, Jr., Thomas P. Bersot, and Robert W. Mahley

From the Gladstone Foundation Laboratories for Cardiovascular Disease, Cardiovascular Research Institute, Departments of Pathology and Medicine, University of California, San Francisco, San Francisco, California 94140

Guido Franceschini and Cesare R. Sirtori

From the Center E. Grossi Paoletti, University of Milan, Milan, Italy

The A-I$_{Milano}$ contains a variant form of apolipoprotein A-I, which unlike normal A-I, has cysteine. We have characterized the A-I$_{Milano}$ isoforms in two affected subjects, D. M. and D. G. In these subjects, the two most prominent A-I$_{Milano}$ isoforms were displaced from the corresponding normal A-I isoforms by a single charge unit toward the anode, as determined by isoelectric focusing. The amino acid compositions of the purified isoforms, which were separated by either preparative isoelectric focusing or thiopropyl-Sepharose 6B chromatography, indicated that the A-I$_{Milano}$ from these subjects contained both a normal A-I (A-I$_N$) and a cysteine-containing variant A-I (A-I$_{Cys}$). Furthermore, amino acid analyses suggested that the A-I$_{Cys}$ differed from normal A-I by a single cysteine for arginine substitution, which was sufficient to account for the charge difference between the two proteins. Partial sequence analysis revealed that an arginine in normal A-I was replaced by cysteine in the variant A-I at residue 173. Consistent with the amino acid analyses, the cysteine-containing isoforms shifted one charge unit toward the cathode after modification of the cysteine residue with cysteamine. Quantitation of the relative amounts of the A-I$_N$ and A-I$_{Cys}$ in D. M. and D. G. by thiopropyl-Sepharose 6B chromatography revealed that the relative levels were different in each subject. The percentage of the total A-I represented by the A-I$_N$ in D. M. and D. G. was 16.1 and 25.7%, respectively. The demonstration of variable amounts of normal A-I in A-I$_{Milano}$ subjects raises some interesting questions regarding the genetics, regulation, and metabolism of apolipoprotein A-I.

The A-I$_{Milano}$[1] contains a variant form of apo-A-I. Thus far, the A-I$_{Milano}$ has been documented in five members of an Italian family (the D. family) (1-3). Affected family members have demonstrated moderately elevated plasma triglyceride levels, nearly normal plasma cholesterol levels, and markedly reduced levels of HDL and apo-A-I (1).

The A-I$_{Milano}$ condition is clinically and biochemically distinct from other familial disorders characterized by decreased HDL levels, including Tangier disease (4-6), lecithin: cholesterol acyltransferase deficiency (7), "fish-eye" disease (8), and cerebrotendinous xanthomatosis (9). Despite low levels of HDL, the A-I$_{Milano}$ subjects have not shown any evidence of premature vascular disease (1). This observation is interesting in light of current epidemiological evidence indicating that vascular disease is inversely correlated with plasma HDL levels (for review, see Ref. 10).

Unlike apo-A-I from normal subjects, the A-I$_{Milano}$ variant contains the amino acid cysteine (2). This variant exists in several forms in the HDL of affected subjects: a M_r = 55,000 disulfide-linked dimer of A-I, a M_r = 35,000 mixed disulfide dimer of apo-A-I and apo-A-II, and a M_r = 28,000 monomer of A-I. The A-I$_{Milano}$ is immunochemically identical with normal A-I and, with the exception of cysteine, its amino acid composition closely resembles that of normal A-I (2).

After demonstrating the A-I$_{Milano}$ variant in one member of the D. family, we studied other family members to examine more closely its structural relationship with normal A-I. The results indicate that affected family members had normal A-I, as well as the cysteine-containing variant, and that a cysteine for arginine substitution occurs in the variant.

MATERIALS AND METHODS

Lipoprotein and Apoprotein Preparation—The HDL (d = 1.063 to 1.21) were prepared from the plasma of fasted subjects by sequential ultracentrifugation in a Ti-60 rotor (Beckman Instruments) at 59,000 rpm. The HDL were dialyzed against distilled H_2O containing 0.01% EDTA, pH 7.4, and then delipidated with $CHCl_3$/MeOH (2:1, v/v) as previously described (11). The apo-HDL were solubilized in 6 M guanidine containing 0.1 M Tris-Cl, 0.01% EDTA, pH 7.4, and applied to a Sephadex G-200 column (295 × 2.5 cm) equilibrated with 4 M guanidine containing 0.1 M Tris-Cl, 0.01% EDTA, pH 7.4. Pooled fractions were exhaustively dialyzed against 5 mM NH_4HCO_3, pH 7.8, and lyophilized. Apoprotein purity was monitored by sodium dodecyl sulfate gel electrophoresis, performed as previously described (12). In some instances, the apo-HDL or isolated disulfide-linked complexes were reduced by incubation with 1.0% (v/v) β-mercaptoethanol for 4 h at 37 °C prior to application to a Sephacryl S-300 column equilibrated with 4 M guanidine containing 0.1 M Tris-Cl, 0.01% EDTA, 0.1% β-mercaptoethanol, pH 7.4.

Isoelectric Focusing—Analytical isoelectric focusing on polyacrylamide gels (6%) was performed over the pH range 4 to 6 (LKB Ampholines), according to the method of Pagnan *et al.* (13), with minor modifications (12). Selected samples were modified with cysteamine (β-mercaptoethylamine, Sigma Chemical) as previously described (11). Control samples were treated in parallel with 0.1% (v/v)

* This work was supported in part by Grant HL27455 from the United States Public Health Service. The costs of publication of this article were defrayed in part by the payment of page charges. This article must therefore be hereby marked "*advertisement*" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

[1] The abbreviations used are: A-I$_{Milano}$, the total complement of A-I from the subjects (mixture of A-I$_N$ and A-I$_{Cys}$); A-I$_N$, the A-I component from the subjects corresponding to normal A-I; A-I$_{Cys}$, the A-I component from the subjects having a cysteine substituted for an arginine at residue 173; apo-, apolipoprotein; HDL, high density lipoproteins.

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β-mercaptoethanol. Preparative isoelectric focusing was performed in a Multiphor flatbed apparatus (LKB Instruments) using Sephadex G-200 as a bed support. Electrophoresis at 8 watts was carried out for 18 h at 4 °C as previously described (11). Isoform bands were located on the bed with the paper print technique (LKB Application Note 198, LKB Instruments, Bromma, Sweden) and eluted from the gel with 4 M guanidine containing 0.1 M Tris-Cl, 0.01% EDTA, pH 7.4. After exhaustive dialysis against 5 mM NH_4HCO_3, the fractions were lyophilized and the residual Ampholines were removed by $(NH_4)_2SO_4$ precipitation of the protein as described (11).

Thiopropyl-Sepharose 6B Chromatography—Lyophilized A-I (7 to 15 mg) obtained by S-300 chromatography of a reduced HDL sample was solubilized in 3 to 4 ml of 6 M guanidine containing 0.1 M Tris-Cl, 0.01% EDTA, 50 mM dithiothreitol, pH 7.4. The mixture was allowed to stand overnight at room temperature under a nitrogen atmosphere and then dialyzed against deoxygenated column buffer (4 M guanidine, 0.1 M Tris-Cl, 0.01% EDTA, pH 7.4) to remove the dithiothreitol. The sample was then applied to a column (10 × 1.0 cm, Kontes Glass) containing 4 g of thiopropyl-Sepharose 6B (Pharmacia Fine Chemicals) equilibrated with column buffer. The sample was recycled through the column three times to ensure complete reaction of all the cysteine-containing components. After elution of the unbound fraction, the bound fraction was eluted with column buffer containing 50 mM dithiothreitol. Pooled fractions were dialyzed against 5 mM NH_4HCO_3 and lyophilized.

Amino Acid Analysis—Samples for analysis were hydrolyzed in 0.45 ml of 6 N HCl for 20 h at 110 °C in sealed, evacuated tubes. The tubes were opened and dried *in vacuo*, and the sample were dissolved in 0.2 N sodium citrate, pH 2.2. Analyses were performed on a Beckman 121MB analyzer equipped with a Model 126 Data System. Multiple analyses represent separate hydrolysates, not repetitive analyses of the same hydrolysate.

Sequence Analysis—Apoprotein A-I$_{Milano}$ (13.4 mg), eluted by Sephadex G-200 chromatography, was lyophilized. The protein was dissolved in 2.5 ml of 70% HCOOH containing 402 mg of CNBr (Pierce) and reacted for 24 h at room temperature. After lyophilization, the reaction mixture was dissolved in 3 ml of 30% HCOOH and fractionated on a Sephadex G-50 (fine beads) column (2.5 × 190 cm) equilibrated in 0.02 N HCl. Resultant fractions were analyzed for cysteine-containing peptides by performic acid oxidation followed by amino acid analysis. The cysteine-containing peptide CB4, corresponding to the COOH-terminal CNBr peptide of apo-A-I, was subjected to automated sequence analysis in the presence of 2 mg of Polybrene on a Beckman 890C Sequencer using a 0.1 M Quadrol program (No. 122974). After conversion in 1 N HCl at 80 °C for 10 min, phenylthiohydantoin amino acids were identified by high performance liquid chromatography as described (11).

RESULTS

The original studies of the A-I$_{Milano}$ were carried out on the apolipoproteins isolated from the HDL of subject D. V., the proband of the D. family. In order to characterize further this variant form of A-I, two other affected family members were selected for study. They were the proband's son, D. M., and brother, D. G.

As previously reported (2), the HDL apoprotein pattern of D. V. on sodium dodecyl sulfate gels differed significantly from that of a normal subject. As shown in Fig. 1, the nonreduced HDL of subject D. V. contained the M_r = 55,000 A-I$_{Milano}$ dimer and the M_r = 35,000 A-I$_{Milano}$-A-II mixed disulfide complex, two forms that are not present in normal HDL. In addition to these bands, apoproteins migrating in positions corresponding to normal A-I and A-II were present. When the HDL were reduced with β-mercaptoethanol, the A-I$_{Milano}$ dimer and mixed disulfide bands disappeared, while there was an increase in the monomer A-I band (M_r = 28,000). The sodium dodecyl sulfate apoprotein patterns of both nonreduced and reduced HDL from D. M. and D. G. were essentially identical with the pattern observed for D. V. (Fig. 1).

The initial isolation of the monomeric A-I of apo-HDL from D. M. and D. G. was performed by gel filtration on Sephadex G-200 following β-mercaptoethanol reduction of the total HDL apo-A-I. The isoelectric focusing pattern of each preparation was determined over the pH range 4 to 6 on polyacrylamide gels. The major isoforms of the A-I$_{Milano}$ did not coelectrophorese with normal A-I but were displaced by what appeared to be a single charge unit toward the anode (Fig. 2). However, a minor band in the position corresponding to the major A-I isoform (pI = 5.7) from normal subjects was also present in the affected subjects. The concentration of this isoform appeared to vary among the three family members, as ascertained qualitatively by the staining intensity of this band. As shown in Fig. 2, the focusing patterns of the isolated A-I preparations were more complex than the patterns of A-I from freshly prepared HDL, *i.e.* the more acidic A-I isoforms were more prominent. The increased prominence of the more acidic isoform bands in A-I following isolation of the apoprotein from normal subjects has been described previously (14–16). It has been suggested that these acidic bands result from partial deamidation of A-I during isolation (14, 15). Even with this added complication, however, the differences in the focusing patterns between the isolated apo-A-I from normal and affected subjects were still apparent.

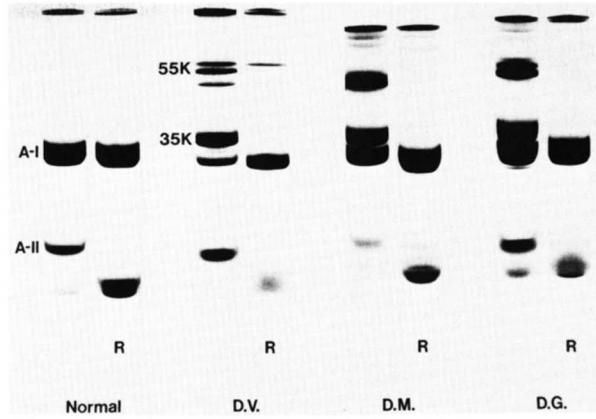


FIG. 1. **Sodium dodecyl sulfate-polyacrylamide gels of control and mercaptoethanol-reduced (*R*) HDL from a normal subject and D. family members.** Lipoprotein protein applied to each gel was 40 μg.

Previously, we have shown that it is possible to detect the presence of cysteine in a protein by monitoring the change in the isoelectric focusing pattern after cysteamine modification (11). The reagent cysteamine reacts with the neutral cysteine residues in a protein via disulfide bond formation and converts each cysteine to a positively charged lysine analogue. The charge on the cysteamine-treated protein is increased by one positive charge unit for each cysteine residue modified. Cysteamine treatment of the A-I$_{Milano}$ from D. M. and D. G. resulted in a shift of focusing patterns toward the cathode, *i.e.* they became more basic. With the exception of the isoform corresponding to the major normal A-I isoform, the A-I$_{Milano}$ isoforms isolated from D. M. and D. G. shifted one position. A shift in isoelectric focusing patterns was also observed for the A-I isoforms in the HDL of affected subjects (Fig. 2). In contrast, the A-I from normal subjects was not affected by cysteamine treatment. The isoform pattern obtained from the HDL of subject D. M. following cysteamine treatment was virtually indistinguishable from the pattern obtained with the A-I of normal HDL (Fig. 2). These results suggested that the major A-I$_{Milano}$ isoforms from these subjects contained 1 residue of cysteine/mol, and that a normal form of A-I (containing no cysteine) was present in the subjects.

To investigate further the possibility that the A-I$_{Milano}$ subjects possessed normal A-I in addition to the variant A-I, we separated the major A-I isoforms of D. M. by preparative

isoelectric focusing. The effect of cysteamine treatment on the focusing patterns for three isoforms of A-I from D. M. was determined (Fig. 3). The most basic isoform (pI = 5.7) of the mixture, which corresponded to the major isoform in normal A-I, was not affected by cysteamine treatment (Fig. 3). This isoform has been designated A-I$_N$ to indicate that it might represent normal A-I. The next band (pI = 5.6), which corresponded to the major isoform in the A-I$_{Milano}$ mixture, partially shifted one charge unit after cysteamine treatment, indicating the presence of both cysteine-containing and noncysteine-containing A-I molecules. This intermediate band, referred to as A-I$_{Int}$, appeared to represent a mixture of A-I$_N$ (pI = 5.6) and a cysteine-containing variant. The third isoform (pI = 5.5) completely shifted one charge unit after cysteamine treatment, indicating that it contained 1 cysteine residue/mol. It was designated A-I$_{Cys}$ to show that it contained cysteine.

The amino acid compositions of normal A-I and the A-I isoforms of D. M. are presented in Table I. These data confirm that the A-I$_{Milano}$ isoforms from D. M. included both cysteine-containing and noncysteine-containing A-I molecules. In addition, there was excellent agreement between the amino acid composition of A-I$_N$ from subject D. M. and normal A-I, including the absence of cysteine and isoleucine. The composition of the isoform intermediate between A-I$_N$ and A-I$_{Cys}$ (A-I$_{Int}$), also agreed well with the composition of normal A-I, with the exception of the occurrence of 0.6 residue of cysteine. These results further suggest that this band represents a mixture of A-I$_N$ and A-I$_{Cys}$. The composition of the A-I$_{Cys}$ isoform differed significantly from normal A-I in cysteine and arginine content, with the A-I$_{Cys}$ containing 1 residue of cysteine and approximately 1 residue of arginine less than normal A-I. These data indicate that A-I$_{Cys}$ may differ from normal A-I by only a single cysteine for arginine substitution.

TABLE I

Amino acid composition of normal A-I and the A-I isoforms of subject D. M.

	Normal A-I[a]	A-I$_N$ (n = 12)	A-I$_{Int}$ (n = 17)	A-I$_{Cys}$ (n = 14)
		residues/mol		
Asp	21	20.9	21.0	21.0
Thr	10	9.7	9.8	9.7
Ser	15	13.9	13.9	13.9
Glu	46	47.5	47.6	47.3
Pro	10	10.3	10.7	10.6
Gly	10	10.5	10.4	10.5
Ala	19	19.0	19.0	18.6
Val	13	13.1	13.0	13.0
Met	3	2.8	2.6	2.7
Ile	0	0	0	0
Leu	37	36.9	36.6	36.9
Tyr	7	6.9	6.8	6.9
Phe	6	6.1	6.2	6.2
Lys	21	20.8	20.6	20.9
His	5	5.0	4.8	4.9
Arg	16	15.6	15.3	14.9
Cys[b]	0	0	0.6	1.0

[a] From Ref. 17.

[b] Determined by performic acid oxidation.

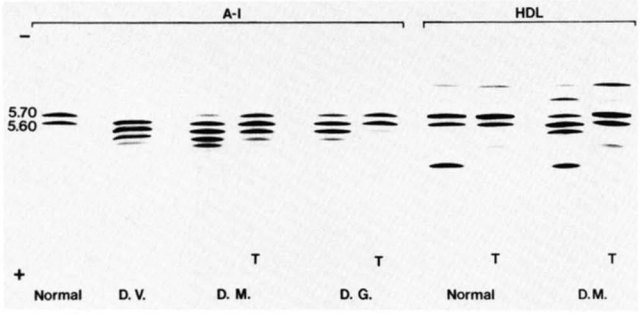


FIG. 2. **Isoelectric focusing on polyacrylamide gels (pH 4 to 6) of apo-A-I from a normal subject and D. family members, and of HDL from a normal subject and subject D. M.** Treated samples (*T*) were modified by incubation with a 0.1% (w/v) cysteamine solution. The amount of A-I or HDL applied to each gel was 20 or 40 μg, respectively. The pI values of the isoforms were determined by measuring the pH of a blank gel at a position corresponding to the position for each isoform.

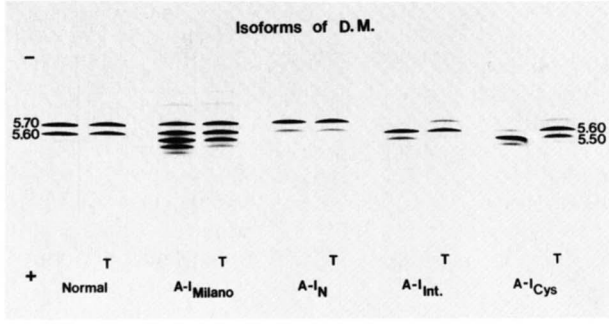


FIG. 3. **Isoelectric focusing on polyacrylamide gels (pH 4 to 6) of the A-I from a normal subject, the A-I$_{Milano}$ from subject D. M., and the isoforms of D. M. isolated by preparative isoelectric focusing.** Treated samples (*T*) were modified by incubation with a 0.1% (w/v) cysteamine solution. The amount of A-I or purified isoforms applied to each gel was 20 or 10 μg, respectively.

Further evidence of this substitution was obtained by partial sequence analysis of the cysteine-containing CNBr peptide. The Sephadex G-50 elution profile of a CNBr digest of A-I$_{Milano}$, which contained both the variant and normal forms, is presented in Fig. 4. The amino acid compositions of the CNBr peptides are given in Table II. The composition of peptide CB4 closely resembled that of the COOH-terminal CNBr peptide of normal A-I (17) except for the presence of cysteine. (The designations used for these peptides are comparable to those used in Ref. 17.) Partial sequence analysis of peptide CB4 (Table III) revealed that cysteine was present at cycle 25, which corresponds to residue 173 in the sequence of normal A-I (17). Also detected at cycle 25 was a small amount of arginine, which was due to the presence of some normal A-I in the A-I$_{Milano}$ preparation used for the sequence analysis.

The demonstration of the presence of normal A-I in the HDL of affected family members necessitated further studies to quantitate the A-I$_N$ in the A-I$_{Milano}$. Due to the presence of a mixture of both cysteine-containing and noncysteine-containing A-I in the isoform band referred to as A-I$_{Int}$ (Fig. 3 and Table I), preparative isoelectric focusing could not be used to separate these two forms of A-I quantitatively. However, it was possible to separate them using thiopropyl-Sepharose 6B chromatography. The monomeric A-I obtained by gel filtration from reduced HDL preparations was used to isolate the A-I$_N$ and A-I$_{Cys}$ isoforms and to determine the levels of each in the total HDL.

The fractions isolated by thiopropyl-Sepharose 6B chromatography were monitored by isoelectric focusing (Fig. 5) and amino acid analysis (Table IV). As demonstrated in Fig. 2, the total A-I isoforms from D. G. shifted only partially after cysteamine treatment; the isoform (pI = 5.7) corresponding to the major isoform in normal A-I did not shift (Fig. 5). The unbound fraction from the thiopropyl column demonstrated several bands, with a major isoform focusing at pI = 5.7. Cysteamine treatment of this fraction did not result in a shift of any of the isoform bands. In contrast, the bound fraction lacked the pI = 5.7 band. Furthermore, all of the isoforms shifted one charge unit after cysteamine treatment. It is important to note that, with the exception of the pI = 5.7

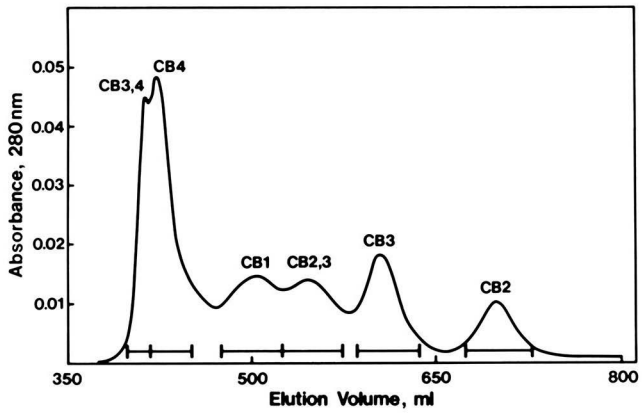


FIG. 4. **Sephadex G-50 elution profile of a CNBr digest of the apo-A-I$_{Milano}$ from subject D. G. (13.4 mg).** The peptide designations correspond to those used by Brewer *et al.* (17). Peptide CB4 was found to contain cysteine and, consequently, was further analyzed (see text).

TABLE II

Amino acid compositions of CNBr peptides of A-I$_{Milano}$ from subject D. G.

Numbers in parentheses are residues for the corresponding peptides of normal A-I (17).

	CB1	CB2	CB3	CB4
	residues/mol			
Asp	11.4 (12)	3.3 (3)	0.3 (0)	6.0 (6)
Thr	4.3 (5)	0.3 (0)	0.4 (0)	4.8 (5)
Ser	6.1 (7)	1.2 (1)	1.3 (1)	5.2 (6)
Glu	15.6 (14)	6.5 (7)	11.8 (12)	14.2 (13)
Pro	3.7 (4)	1.2 (1)	1.9 (2)	3.0 (3)
Gly	5.3 (5)	0.5 (0)	2.1 (2)	3.1 (3)
Ala	2.4 (2)	1.2 (1)	2.0 (2)	14.2 (14)
Cys[a]	(0)	(0)	(0)	0.7 (0)
Val	6.2 (7)	2.2 (2)	1.2 (1)	3.2 (3)
Met[b]	+ (1)	+ (1)	+ (1)	(0)
Ile	(0)	(0)	(0)	(0)
Leu	11.1 (11)	2.4 (2)	6.5 (7)	17.0 (17)
Tyr	2.1 (2)	0.8 (1)	0.9 (1)	2.7 (3)
Phe	3.2 (3)	0.8 (1)	0.2 (0)	1.9 (2)
Lys	6.8 (6)	4.7 (5)	3.0 (3)	6.7 (7)
His	(0)	(0)	0.8 (1)	4.0 (4)
Arg	3.6 (4)	0.2 (0)	2.6 (3)	8.1 (9)
Yield %	20	50	61	34

[a] Determined by performic acid oxidation.
[b] Present as homoserine lactone.

isoform, several of the isoforms of nonbound and bound fractions had identical pI values. As discussed earlier, the more acidic isoforms appeared to have been generated during isolation. The amino acid composition of the unbound fraction of A-I from subject D. M. demonstrated that this fraction was essentially identical to normal A-I and lacked cysteine (Table IV). The bound fraction contained 1 residue of cysteine and approximately 1 residue of arginine less than normal A-I. The unbound fraction, representing A-I$_N$ from the thiopropyl-Sepharose 6B column, accounted for 16.1 and 25.7% of the total A-I from subjects D. M. and D. G., respectively. The protein in the two fractions represented more than 90% of the total protein applied to the column.

In addition to apo-A-I, apo-A-II was isolated from these subjects, either from nonreduced HDL or from the reduction of the A-I-A-II dimer. The amino acid composition of D. M. A-II isolated from the mixed disulfide was compared to that of normal A-II (Table V). The composition of A-II from D. M. was essentially identical with that of A-II from normal subjects, including the lack of significant levels of arginine and histidine in the protein. These data indicate that the A-II in this family was probably normal. Previously, we demonstrated that the A-II from D. V. focused with a pI identical with

TABLE III

Partial sequence analysis of peptide CB4 of A-I$_{Milano}$ from subject D. G.

Residue No.	Cycle No.	Amino acid identified	nmol	$n + 1/n$
	0		110	
149	1[a]	Arg	7.2	
150	2	Asp	25.0	0.20
151	3[a]	Arg	5.2	
152	4	Ala	75.6	0.12
153	5[a]	Arg	4.4	
154	6	Ala	79.2	0.17
155	7[a]	His	3.9	
156	8	Val	69.3	0.15
157	9	Asp	20.7	0.23
158	10	Ala	67.8	0.20
159	11	Leu	62.7	0.28
160	12[a]	Arg	3.7	
161	13	Thr	41.0	0.18
162	14[a]	His	2.8	
163	15	Leu	73.0	0.31
164	16	Ala	60.1	0.24
165	17[b]	Pro	30.1	0.18
166	18	Tyr	24.2	0.24
167	19	Ser	26.9	0.26
168	20	Asp	12.3	0.11
169	21	Glu	23.4	0.44
170	22	Leu	56.2	0.32
171	23[a]	Arg	2.7	0.16
172	24	Gln	6.4	0.45
173	25[c]	Cys	7.5	0.28
174	26	Leu	33.5	0.56
175	27	Ala	34.8	
176	28	Ala	41.5	0.35
177	29[a]	Arg	1.7	
178	30	Leu	35.2	0.56
179	31	Glu	12.4	0.40
180	32	Ala	29.8	0.44
181	33	Leu	36.0	0.44
182	34	Lys	8.0	0.89
183	35	Glu	11.3	0.45
184	36	Asn	0.9	

[a] From aqueous phase analysis. Yields of the charged amino acids Arg and His, and sometimes Asp and Glu, are dependent on the age of the column. Yields are lower in aged columns where free silanol groups can irreversibly adsorb these amino acid derivatives.
[b] Double cleavage at this cycle.
[c] Arginine, 0.5 nmol, was also detected.

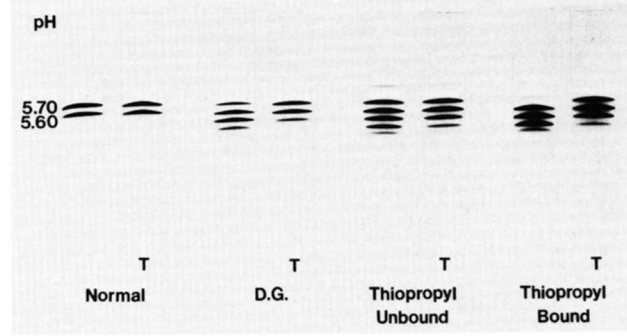


FIG. 5. **Isoelectric focusing on polyacrylamide gels (pH 4 to 6) of A-I from a normal subject and D. G., and unbound and bound fractions of D. G. A-I from thiopropyl-Sepharose 6B chromatography.** Treated samples (*T*) were modified by incubation with 0.1% (w/v) cysteamine solution. The amount of protein applied to each gel was 25 μg.

TABLE IV

Amino acid composition of normal A-I and the thiopropyl-Sepharose fractions of D. M. A-I

	Normal A-I[a]	Unbound (A-I$_N$) (n = 3)	Bound (A-I$_{Cys}$) (n = 3)
		residues/mol	
Asp	21	21.2	21.2
Thr	10	9.9	9.9
Ser	15	13.7	13.4
Glu	46	47.8	47.6
Pro	10	10.6	10.6
Gly	10	10.7	10.6
Ala	19	18.8	18.3
Val	13	12.9	12.9
Met	3	2.8	2.9
Ile	0	0	0
Leu	37	36.7	37.1
Tyr	7	6.7	6.8
Phe	6	5.6	5.9
Lys	21	20.7	20.7
His	5	4.9	4.9
Arg	16	16.0	15.2
Cys[b]	0	0	1.1

[a] From Ref. 17.
[b] Determined by performic acid oxidation.

TABLE V

Amino acid composition of normal A-II and A-II obtained from D. M.

	Normal A-II[a]	D. M. A-II (n = 8)
	residues/mol	
Asp	3	3.3
Thr	6	5.8
Ser	6	5.6
Glu	16	16.1
Pro	4	4.2
Gly	3	3.2
Ala	5	5.0
Val	6	6.1
Met	1	1.0
Ile	1	1.0
Leu	8	8.3
Tyr	4	3.8
Phe	4	3.9
Lys	9	8.5
His	0	<0.1
Arg	0	0.3
Cys[b]	1	1.0

[a] From Ref. 18.
[b] Determined by performic acid oxidation.

normal A-II (2), as was also observed for the A-II from D. G. and D. M. (Fig. 2).

DISCUSSION

In this study, two additional affected members of the D. family, a son (D. M.) and a brother (D. G.) of the proband (D. V.), were selected for further characterization of the A-I$_{Milano}$. On the basis of the abnormal apoprotein profiles obtained on acrylamide gels, it was indicated that the variant A-I was present in the HDL of these two subjects. These apoprotein patterns were identical with the D. V. pattern, which included the A-I$_{Milano}$ dimer and the A-I$_{Milano}$-A-II mixed disulfide. Monomeric A-I and A-II were also present. In addition, as previously described for D. V. (2), the major A-I$_{Milano}$ isoforms of D. M. and D. G. did not co-electrophorese with the major normal A-I isoforms, but were displaced by one charge unit toward the anode upon isoelectric focusing. This pattern could be made to resemble the normal pattern by treating the A-I$_{Milano}$ with cysteamine. The results suggest that the major A-I$_{Milano}$ isoforms in these subjects contained 1 residue of cysteine/mol and that the cysteine residue of the variant was substituted for a basic residue in the normal A-I. A cysteine for arginine substitution was indicated by amino acid compositional data obtained from the isoforms. This substitution was confirmed by a partial sequence analysis of the A-I$_{Milano}$, showing a cysteine residue substituted for an arginine at residue 173. Therefore, this particular variant of A-I could be designated as apo-A-I$_{Cys\ 173}$. This single amino acid interchange was sufficient to account for the charge difference between the variant and normal forms of A-I.

In the original characterization studies of subject D. V., we reported the presence of isoleucine and 2 mol of cysteine in the A-I$_{Milano}$ (2). The A-I$_{Milano}$ isolated from D. M. and D. G. in this study did not contain isoleucine, either in the A-I fractions from gel filtration and thiopropyl-Sepharose columns or in the isoforms separated by preparative isoelectric focusing. Repetitive cysteine determinations and cysteamine treatment have now established that there is only 1 residue of cysteine/mol of protein in the cysteine-containing A-I variant. The apparent discrepancy regarding isoleucine may be explained by an isoleucine-containing contaminant in the original studies of D. V., or by the presence of the isoleucine in a minor A-I isoform of D. V. Other laboratories have detected varying amounts of isoleucine in A-I preparations (19, 20), and isoleucine has been found to be associated with the most acidic minor A-I isoforms (16).

In addition to A-I$_{Cys}$, the A-I$_{Milano}$ in these subjects contained normal A-I (A-I$_N$). The A-I$_{Milano}$ isoform with a pI corresponding to the main isoform of normal A-I was shown by amino acid analysis and cysteamine treatment to represent normal A-I in these subjects. Previously, it had been demonstrated that the staining intensity of this band on isoelectric focusing gels (relative to the main A-I$_{Milano}$ isoforms) varied among the affected family members, and that subject D. V. had the lowest concentration (3). In this study, we have taken advantage of the fact that the variant A-I contained cysteine, and have quantitated the amount of normal A-I in these subjects using thiopropyl-Sepharose 6B chromatography. The proportion of A-I$_N$ in the HDL of D. M. and D. G. was 16.1 and 25.7%, respectively. Thus, at least three-fourths of the A-I$_{Milano}$ was represented by the A-I$_{Cys}$.

Recently Utermann *et al.* (21) suggested that the A-I$_{Milano}$ defect represents a heterozygous condition characterized by the presence of the variant and normal A-I. They reported that one of the A-I$_{Milano}$ isoforms appeared to be normal A-I. This suggestion was based on the observations that this isoform was not affected by reducing agents on two-dimensional gels and that the concentration of this isoform was not changed by reduction. However, no amino acid analysis was given to support the designation of this isoform as normal A-I. Utermann *et al.* (21) viewed their results as a contradiction of our original studies on subject D. V., where we reported little, if any, of this isoform in the HDL of D. V. (2). However, as determined in our present study, the proportion of normal A-I in these subjects varied, ranging from 16.1 to 25.7%.

The existence of both normal A-I and the A-I$_{Cys}$ variant in these subjects has very interesting implications regarding the genetics, regulation, and metabolism of apo-A-I. The fact that the two subjects in this study had variable amounts of A-I$_N$, but that the amount was always less than half of the total A-I, suggests the possibility that A-I$_N$ and A-I$_{Cys}$ may not be allelic proteins and that the subjects were not simple heterozygotes. Regardless of whether A-I$_N$ and A-I$_{Cys}$ are allelic, other factors must be involved to account for quantitative differences in the concentrations of A-I$_{Cys}$ and A-I$_N$ among the subjects, especially the low concentrations of A-I$_N$ relative to A-I$_{Cys}$. Such factors include the possibility of multiple A-I genes, differential gene regulation for the synthesis of A-I$_N$

and A-I_{Cys}, or different metabolic behavior of A-I_N and A-I_{Cys}, *e.g.* different sites or rates of catabolism. Further studies are required to distinguish among these possibilities.

Acknowledgments—We wish to thank David Begert, Daniel Goodman, and Jana Seymour for their excellent technical assistance, and Reed Harris for aid in amino acid and sequence analysis. We acknowledge Russell Levine for editorial assistance, Richard Wolfe for manuscript preparation, and Gwen Watson for graphic arts.

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